

AWS SECURE CLOUD ARCHITECTURE PROJECT

Architecture, Networking, IAM, Monitoring & Validation

Document Type	Technical Project Documentation
Environment	AWS / us-east-1
Status	Training Lab / Portfolio Draft
Primary Scope	Secure 3-VPC web application with centralized routing and monitoring

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1. Executive Summary

This project implements a segmented AWS cloud environment for a PHP-based web application. The architecture separates the web, testing, and database workloads into three VPCs and uses an AWS Transit Gateway as the private connectivity hub. The web application runs on Amazon EC2 with Apache and PHP, while the database tier runs on Amazon RDS for MySQL with public access disabled.

Operational visibility is provided by Amazon CloudWatch and notification delivery by Amazon SNS. The environment was validated through real tests, including a high-CPU event on the web server and a sustained high-connection event on the RDS database. Both tests transitioned the relevant alarms to ALARM and generated SNS email notifications.

2. Objectives and Scope

- Implement role-based access control with separate Admin, Developer, and Tester permissions.
- Build three segmented VPCs for Web, Testing, and Database workloads.
- Use Transit Gateway to provide private connectivity between the VPCs.
- Host the web application on EC2 with Apache/PHP and serve it over HTTPS.
- Keep the RDS database private and expose it only through controlled private connectivity on TCP 3306.
- Implement CloudWatch alarms for EC2 and RDS health signals and send notifications through SNS.
- Validate the monitoring design using repeatable real-world tests.
- Capture the architecture in a professional diagram and GitHub-ready documentation.

3. High-Level Architecture

The active environment contains three VPCs, each with a distinct purpose. Web traffic enters through an Internet Gateway and reaches the Website EC2 instance in a public subnet. Private inter-VPC traffic is routed through TGW-01. The Database VPC contains the RDS MySQL service with public access disabled. Testing resources are isolated in a dedicated Testing VPC.

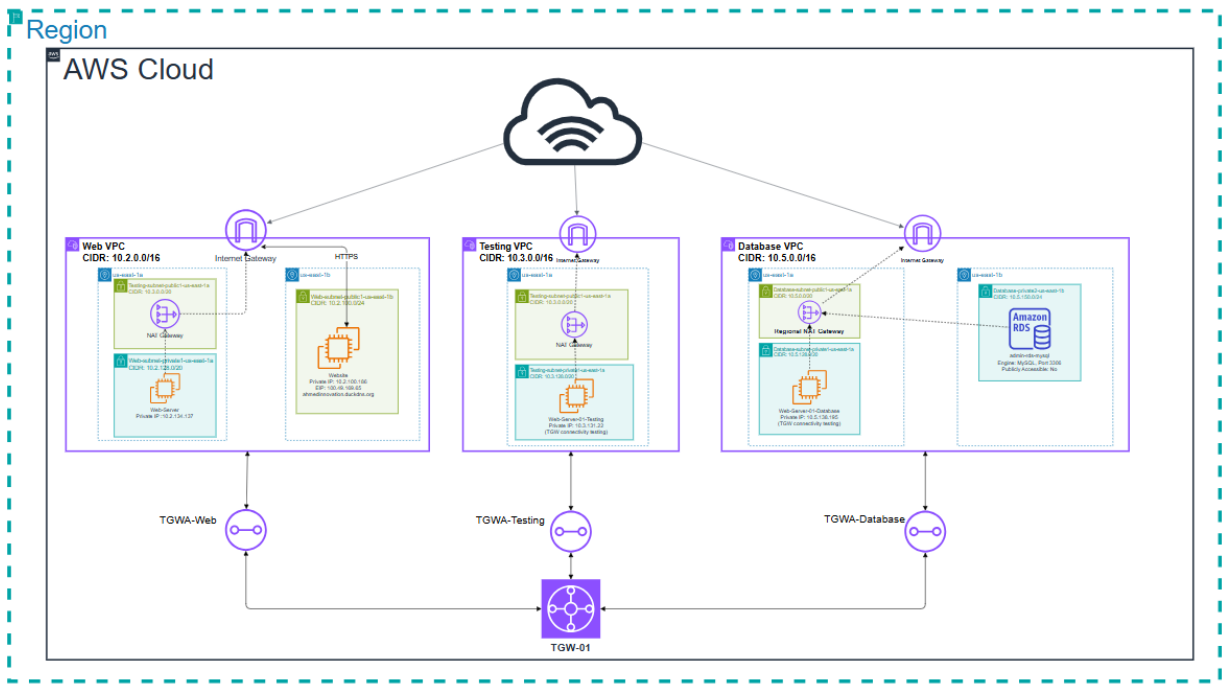


Figure 1. Current architecture diagram draft from the training environment. Final polished diagram should be validated against the AWS Console before GitHub publication.

Layer	AWS components	Purpose
Edge / Internet	Internet, Internet Gateways, DuckDNS, HTTPS	Public entry path for the web workload.
Web	Web VPC, EC2 Website, Bastion Host, public/private subnets, NAT routing	Hosts Apache/PHP and provides controlled private access.
Inter-VPC	Transit Gateway TGW-01	Provides private routing between Web, Testing and Database VPCs.
Testing	Testing VPC, Testing EC2	Connectivity and network validation environment.
Database	Database VPC, RDS MySQL, Database EC2	Private data tier and TGW connectivity validation.
Operations	CloudWatch + SNS	Monitoring, alarms and email notification.
Governance	IAM	Role-based permissions and least-privilege access model.

4. Network Architecture

4.1 VPCs

VPC	CIDR	Purpose	DNS
Web VPC	10.2.0.0/16	Website and bastion access layer	Resolution + hostnames enabled
Testing VPC	10.3.0.0/16	TGW and connectivity validation	Resolution + hostnames enabled
Database VPC	10.5.0.0/16	Private database tier	Resolution + hostnames enabled

4.2 Subnet layout

VPC / AZ	Subnet	CIDR	Type	Workload
Web / us-east-1a	Web-subnet-public1-us-east-1a	10.2.0.0/20	Public	No listed workload
Web / us-east-1a	Web-subnet-private1-us-east-1a	10.2.128.0/20	Private	Bastion Host
Web / us-east-1b	Web-subnet-public1-us-east-1b	10.2.100.0/24	Public	Website EC2
Testing / us-east-1a	Testing-subnet-public1-us-east-1a	10.3.0.0/20	Public	No listed workload
Testing / us-east-1a	Testing-subnet-private1-us-east-1a	10.3.128.0/20	Private	Testing EC2
Database / us-east-1a	Database-subnet-public1-us-east-1a	10.5.0.0/20	Public	No listed application workload
Database / us-east-1a	Database-subnet-private1-us-east-1a	10.5.128.0/20	Private	Database EC2
Database / us-east-1b	Database-private2-us-east-1b	10.5.150.0/24	Private	RDS subnet-group member

4.3 Transit Gateway

TGW-01 (tgw-000b060df9c1c1e1a) is the central routing hub. Its route table has active propagated routes for the three VPC CIDRs.

Destination	Attachment	VPC
10.2.0.0/16	TGWA-Web	Web VPC
10.3.0.0/16	TGWA-Testing	Testing VPC
10.5.0.0/16	TGWA-Database	Database VPC

4.4 Key route patterns

Web private: 0.0.0.0/0 → NAT Gateway | 10.3.0.0/16 → TGW | 10.5.0.0/16 → TGW

Testing private: 0.0.0.0/0 → NAT Gateway | 10.2.0.0/16 → TGW | 10.5.0.0/16 → TGW

Database private (us-east-1b): 0.0.0.0/0 → NAT Gateway | 10.2.0.0/16 → TGW | 10.3.0.0/16 → TGW

5. Compute and Application Layer

Instance	VPC / Subnet	Private IP	Public IP	Purpose
Website	Web / 10.2.100.0/24	10.2.100.186	100.49.169.65	Static website, Apache/PHP, RDS connectivity
Web-Server	Web / 10.2.128.0/20	10.2.134.137	None	Private access to private servers
Web-Server-01-Testing	Testing / 10.3.128.0/20	10.3.131.22	None	TGW connectivity testing
Web-Server-01-Database	Database / 10.5.128.0/20	10.5.138.195	None	TGW connectivity testing

5.1 Web software stack

Item	Observed value
Operating system	Amazon Linux 2023
Web server	Apache 2.4.68
PHP	PHP 8.5.10
Domain	https://ahmedinnovation.duckdns.org
HTTPS	Working; HTTP requests return 200 OK over verified HTTPS
Certificate	Let's Encrypt certificate for the DuckDNS hostname

6. Database Layer

The data tier uses Amazon RDS for MySQL in the Database VPC. The RDS instance is not publicly accessible and listens on TCP 3306. Connectivity was verified from the Web environment using the RDS endpoint.

Property	Value
DB identifier	admin-rds-mysql
Engine	MySQL Community
Port	3306
Public access	Disabled
VPC	Database VPC / 10.5.0.0/16
Security group	SG-RDS-Database
Application database	website_db
Application table	visitors
Validated records	2 rows observed during the lab

Database validation

A MySQL query from the Web EC2 successfully returned the available databases, then selected `website\_db` and confirmed the `visitors` table with 2 current records.

6.1 RDS subnet group

```
default-vpc-05a4cc404c0692656
```

```
Members: 10.5.0.0/20, 10.5.128.0/20, 10.5.150.0/24
```

The inventory identifies these as the subnet-group members; the diagram should not imply that a single RDS instance endpoint resides in one specific subnet unless the console explicitly shows that placement.

7. Identity and Access Management

IAM was used to separate administrative, development, and testing responsibilities. The design demonstrates role-based access control and the principle of least privilege for the Developer and Tester roles.

Role	Permissions
Admin	Full AWS access: Action * on Resource *.
Developer	EC2 StartInstances, StopInstances, RebootInstances, TerminateInstances for the assigned instance; EC2 DescribeInstances/DescribeInstanceStatus/DescribeTags; S3 ListBucket; S3 GetObject/PutObject/DeleteObject for the assigned bucket.
Tester	EC2 DescribeInstances, DescribeInstanceStatus, DescribeTags (read-only observation).

8. Security Controls

Control	Implementation
Network segmentation	Separate Web, Testing and Database VPCs.
Private database	RDS publicly accessible: No.
Network routing	Inter-VPC routes use Transit Gateway rather than public routing.
Security Groups	Dedicated groups for Web, Testing, Database EC2 and RDS.
RDS exposure	Inbound MySQL/TCP 3306 is controlled through SG-RDS-Database.
Transport security	Website served over HTTPS using a trusted certificate.
IAM	Role-based separation between Admin, Developer and Tester.
Secrets hygiene	Credentials excluded from documentation and GitHub.

9. Monitoring and Alerting

CloudWatch alarms were created for both the EC2 Web Server and the RDS database. All alarms publish their ALARM state notification to the SNS topic `CloudInfrastructure-Alerts`.

Alarm	Metric	Condition	Period / Statistic	Test status
EC2-Website-HighCPU	CPUUtilization	> 70%	5 min / Average	Tested: ALARM + SNS email
RDS-admin-mysql-HighCPU	CPUUtilization	> 70%	5 min / Average	Configured; no separate failure test recorded
RDS-admin-mysql-LowStorage	FreeStorageSpace	< 2 GiB (2147483648 bytes)	5 min / Average	Configured; threshold direction corrected; not force-tested
RDS-admin-mysql-HighConnections	DatabaseConnections	> 50	5 min / Average	Tested: ALARM + SNS email

9.1 SNS notification path

EC2/RDS metric → CloudWatch Alarm → SNS (CloudInfrastructure-Alerts) → Email notification

10. Testing and Validation

10.1 EC2 high-CPU test

A controlled CPU load was generated on the Website EC2 using stress-ng. The CloudWatch alarm entered ALARM when the measured CPU utilization reached approximately 99.67%, exceeding the 70% threshold. The SNS notification email confirmed an OK → ALARM state transition.

Observed alarm event: CPU ≈ 99.67% > 70% → EC2-Website-HighCPU → SNS → Email

10.2 RDS high-connections test

Controlled MySQL sessions were opened from the Web EC2 to the RDS endpoint. The observed `Threads\_connected` value reached 63 at the client, while CloudWatch recorded an average `DatabaseConnections` value of about 60.25. This crossed the configured threshold of 50 and triggered the RDS High Connections alarm and SNS email notification.

Observed alarm event: DatabaseConnections ≈ 60.25 > 50 → RDS-admin-mysql-HighConnections → SNS → Email

After the test, the additional sessions were terminated and the connection count returned to 3, confirming recovery to a normal state.

10.3 HTTPS validation

`curl -I https://ahmedinnovation.duckdns.org → HTTP/1.1 200 OK`

10.4 Transit Gateway validation

Connectivity was validated between the Web-side private host and the Testing and Database VPC test instances: 10.2.134.137 → 10.3.131.22 and 10.2.134.137 → 10.5.138.195.

11. HTTPS, DNS and Web Access

Item	Current state
Public hostname	ahmedinnovation.duckdns.org
DNS	Resolves to Elastic IP 100.49.169.65
HTTPS	443 / working
HTTP response	200 OK over HTTPS
Apache SSL configuration	Uses Let's Encrypt fullchain.pem and privkey.pem for the DuckDNS hostname
Application response	Apache + PHP serve the website successfully

12. Architecture Decisions and Notes

- Transit Gateway is the central private connectivity hub between the three VPCs.
- RDS remains private and is not directly Internet-facing.
- The Website EC2 is in the Web public subnet and is the public application entry point.
- The Testing VPC provides a dedicated validation environment rather than production traffic.
- CloudWatch and SNS provide centralized operational visibility and notification.
- The current diagram deliberately excludes the stale 10.4.0.0/16 route and any unlisted services.
- NAT Gateway placement should be taken from the console if the final diagram needs exact NAT topology.

13. GitHub Documentation Structure

```
aws-secure-cloud-architecture/  
├─ README.md  
├─ docs/  
│   ├── architecture.md  
│   ├── network.md  
│   ├── iam.md  
│   ├── security.md  
│   ├── monitoring.md  
│   └─ testing.md  
├─ diagrams/  
│   ├── architecture.drawio  
│   ├── architecture.png  
│   └─ architecture.pdf  
├─ screenshots/  
│   ├── iam/  
│   ├── networking/  
│   ├── rds/  
│   └─ monitoring/  
└─ policies/  
    ├── developer-policy.json  
    └─ tester-policy.json
```

15. Final Validation Checklist

- Web VPC, Testing VPC and Database VPC CIDRs match the active environment.
- Website EC2 is shown in 10.2.100.0/24 and serves HTTPS.
- Bastion Host is shown in 10.2.128.0/20 and has no public IP.
- Testing EC2 is shown in 10.3.128.0/20 and has no public IP.
- Database EC2 is shown in 10.5.128.0/20 and has no public IP.
- Database-private2-us-east-1b (10.5.150.0/24) is shown as PRIVATE.
- RDS admin-rds-mysql is private and uses TCP 3306.
- TGW route table includes only 10.2.0.0/16, 10.3.0.0/16 and 10.5.0.0/16 as active propagated routes.
- CloudWatch alarms and SNS notification path are documented.
- EC2 High CPU and RDS High Connections tests are documented with observed results.
- Low Storage threshold is documented as < 2 GiB and not claimed as force tested.
- The stale 10.4.0.0/16 route is excluded from the active architecture.
- No credentials or secret values are present in the document or GitHub repository.
- Final architecture diagram has been reviewed against the AWS Console.